Henrik Standnes Grønlund

The title of your project thesis should be written here

Any undertitle is written here

Project thesis in Physics and Mathematics
Supervisor: Mats Ernström
Co-supervisor: Douglas S. Seth
December 2026

Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Mathematics

NTNU

ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.
Plan for: 29.09

- Se på hvordan ikdv fungerer og lage en generell klasse for å finne symbolene
- Lage en egen LSM solver som tar inn ferdig lagde bølge matriser og finner symbolene
- Lage mer komplekse sammensatte bølger og se hvordan de oppfører seg i sovleren
-

PREFACE

Throughout the text I assume that the reader is familiar with theory within PDE's, numerical methods and the Fourier-space. This is not intended to be a text anyone can pick up, but some of my fellow students working in similar fields should understand it. (Think Vemund Trym (x2) and Magnus)
You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

CONTENTS

Abstract i

Preface ii

Contents iii

List of Figures iii

List of Tables iv

Abbreviations vi

1 Introduction 1
1.1 Motivation 1
1.2 Project description 2
1.2.1 Stakeholders 2

2 Theory 3

3 Methods 5
3.1 Method 5
3.1.1 Integrating Factor 5
3.1.2 Exponentital Time Differention 5
3.1.3 Split Step 5

4 Results 7

5 Discussion 9
5.1 Future work 9

6 Conclusions 11

References 13

Appendices: 15

A - Github repository 16

B - Sidenote statistics 17

LIST OF FIGURES

B.1 Data set sizes 19

LIST OF TABLES

B.1 Statistics on something 17
B.2 Statistics on something else 18

ABBREVIATIONS

CHAPTER

ONE

INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.1 Motivation

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

1.2 Project description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.2.1 Stakeholders

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

CHAPTER

TWO

THEORY

CHAPTER

THREE

METHODS

3.1 Method

In this section the various methods utilized are presented.

3.1.1 Integrating Factor

For the general PDE, $u_t = \mathcal{L}u + \mathcal{N}(u,t)$, multiplication by the exponetial e^{-Lt} reduces the problem to:

$$v_t = e^{-Lt}\mathcal{N}\left(e^{Lt}v\right)$$
$$v = e^{-Lt}u$$

For purely linear problems this method gives an explicit solution

$$u(x,t) = u(x,0)\,e^{Lt}, \tag{3.1}$$

That can be effectivly computed using matrix products.

In the general case with *Nonlinearities* one can use Runge-Kutta or other time-stepping methods. Based on the paper from Kassam and Trefethen [1], ETD methods seems to work quite well.

3.1.2 Exponentital Time Differention

By integrating the IF formula over a *single* time step h we get

$$u_{n+1} = e^{Lh}u_n + e^{Lh}\int_0^h e^{-L\tau}\mathcal{N}\left(u(t_n+\tau), t_n+\tau\right). \tag{3.2}$$

An exact solution! All that is left is to approximate the integral (see Cox and Matthews)

3.1.3 Split Step

Solving a mixed problem is hard, however you can just solve them separate by taking smaller steps. Using the symmetric split step method you get a second

order solution in time, and it it quite simple to implement using the integrating factor for the linear part, and RK4 or some other solver for the nonlinear part. Then one can combine the solutions. Take a half using the linear method, solve the nonlinearity and then take another halfstep.

CHAPTER

FOUR

RESULTS

CHAPTER

FIVE

DISCUSSION

Discuss your results here.

5.1 Future work

Include a section about what should or could be done in future research, or explain any recommended next steps based on the results you got. This should be the last section in the discussion.

CHAPTER

SIX

CONCLUSIONS

Give a concise summary of your research and finding here, and include a short summary of any future work as well.

REFERENCES

[1] Aly-Khan Kassam and LLoyd N. Trefethen. "Fourth-order time-stepping for stiff PDEs". In: *SIAM j. Sci. Comput* 26.4 (2005).

APPENDICES

A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

Github repository link

- `https://github.com/ninasalvesen/thesis_latex_template`

B - SIDENOTE STATISTICS

B1 - Some random table

Remember to only include one thing per page in the appendices.

Statistic	One	Two
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

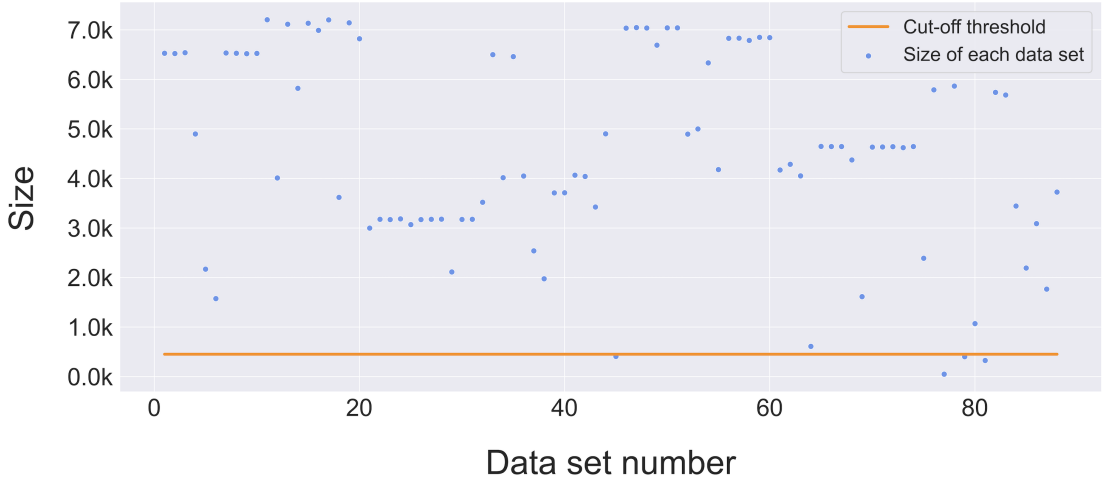
Table B.1: Table of statistics on some sidenote data.

B2 - Some other random table

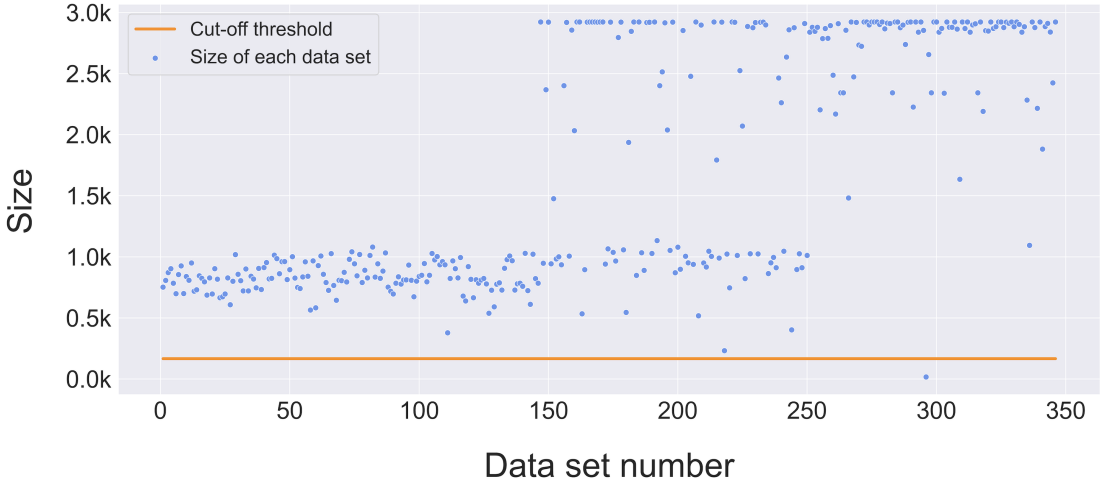
Statistic	Three	Four
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

Table B.2: Table of statistics on some other sidenote data.

B3 - Some random figure



(a) Data set sizes for data X.



(b) Data set sizes for data Y.

Figure B.1: The figures show the data set sizes of X and Y and the proposed cut-off threshold at 10 % of the mean set size.